

```
import numpy as np
```

```
a1=np.array([1,2,3,4,5])  
print("1-dim array",a1,sep='\n',end='\n\n')
```

```
1-dim array  
[1 2 3 4 5]
```

```
a2=np.array([[1,2,4],[4,6,2]])  
print("2-dim array",a2,sep='\n',end='\n\n')
```

```
2-dim array  
[[1 2 4]  
 [4 6 2]]
```

```
a3=np.array([[[1,2,3],[2,3,3],[5,7,1]]])  
print("3-dim array",a3,sep='\n',end='\n\n')
```

```
3-dim array  
[[[1 2 3]  
  [2 3 3]  
  [5 7 1]]]
```

```
zeroes=np.zeros((5,4))  
print(zeroes)
```

```
[[0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]  
 [0. 0. 0. 0.]]
```

```
one=np.ones((5,2))  
print(one)
```

```
[[1. 1.]  
 [1. 1.]  
 [1. 1.]  
 [1. 1.]  
 [1. 1.]]
```

```
identity=np.eye(5)  
print(identity)
```

```
[[1. 0. 0. 0. 0.]  
 [0. 1. 0. 0. 0.]  
 [0. 0. 1. 0. 0.]  
 [0. 0. 0. 1. 0.]  
 [0. 0. 0. 0. 1.]]
```

```
full_array=np.full((5,3),'sreeja')  
print(full_array)
```

```
['sreeja' 'sreeja' 'sreeja']  
['sreeja' 'sreeja' 'sreeja']  
['sreeja' 'sreeja' 'sreeja']  
['sreeja' 'sreeja' 'sreeja']  
['sreeja' 'sreeja' 'sreeja']
```

```
range_arr=np.arange(1,11,1)  
print(range_arr)
```

```
[ 1  2  3  4  5  6  7  8  9 10]
```

```
line_space=np.linspace(1,100,10)  
print(line_space)
```

```
[ 1.  12.  23.  34.  45.  56.  67.  78.  89. 100.]
```

```
rand_int=np.random.randint(30,60)  
print(rand_int)
```

```
48
```

```
rand_float=np.random.rand(3,3)
print(rand_float)
```

```
[[0.11227723 0.23284444 0.68870405]
 [0.78033217 0.95786582 0.54306916]
 [0.44928674 0.30239217 0.62439134]]
```

```
rand_int=np.random.randint(2,9,(3,3))
print(rand_int)
```

```
[[6 4 2]
 [7 6 6]
 [3 2 8]]
```

```
rand_choice=np.random.choice(['sreeja','hannah','allie','grace','sabrina'])
print(rand_choice)
```

hannah

```
np.random.seed(10)
rand_seed=np.random.randint(6)
print(rand_seed)
```

1

```
arr=np.array([[1,2],[3,4],[5,7]])
print(arr.shape)
```

(3, 2)

```
reshape=arr.reshape(1,6)
print(reshape)
```

```
[[1 2 3 4 5 7]]
```

```
arr=np.array([[1,2],[3,4],[5,7]])
flatten=arr.flatten()
print(flatten)
```

```
[1 2 3 4 5 7]
```

```
transpose=arr.T
print(transpose)
```

```
[[1 3 5]
 [2 4 7]]
```

```
arr=np.array([10,45,67,23])
print(arr[0])
print(arr[1])
print(arr[2])
print(arr[3])
print(arr[-1])
print(arr[0:3])
```

```
10
45
67
23
23
[10 45 67]
```

```
arr=np.array([[1,2,3],[3,4,6],[6,8,4]])
print(arr[0,2])
print(arr[0:3,0:2])
print(arr[0:3,1])
```

```
3
[[1 2]
 [3 4]
 [6 8]]
[2 4 8]
```

```
arr=np.array([10,45,67,23])
print(arr+1)
print(arr*0)
print(arr**2)
```

```
print(arr/2)
print(np.sqrt(arr))
```

```
[11 46 68 24]
[0 0 0 0]
[ 100 2025 4489 529]
[ 5.  22.5 33.5 11.5]
[3.16227766 6.70820393 8.18535277 4.79583152]
```

```
print(np.sum(arr))
print(np.mean(arr))
print(np.median(arr))
```

```
145
36.25
34.0
```

```
print(np.mean(arr))
print(np.var(arr))
print(np.std(arr))
```

```
36.25
471.6875
21.718367802392518
```

```
arr=np.array([1,3,4,6,2])
print(np.min(arr))
print(np.max(arr))
```

```
1
6
```

```
print(np.cumsum(arr))
print(np.cumprod(arr))
```

```
[ 1  4  8 14 16]
[ 1  3 12 72 144]
```

```
print(arr>3)
print(arr[arr>3])
```

```
[False False  True  True False]
[4 6]
```

```
print(np.sort(arr))
print(np.unique(arr))
```

```
[1 2 3 4 6]
[1 2 3 4 6]
```